

Generative AI, Embeddings, and RAG

Custom knowledge-base material for the AI & Data Science RAG Chatbot project.

1. Generative AI and LLMs

Generative AI creates new content such as text, code, images, or audio. A Large Language Model processes token sequences and predicts likely next tokens. It can summarize, explain, transform, and generate text but may produce incorrect information.

2. Transformers and Attention

Transformers process sequences using attention mechanisms. Attention allows a token to weigh the relevance of other tokens in the context. Self-attention helps the model represent relationships between words or tokens.

3. Tokens and Prompt Engineering

A token is a unit processed by a language model; it may be a word, part of a word, punctuation mark, or character sequence. Prompt engineering means writing instructions and context clearly. Zero-shot prompting gives no examples; few-shot prompting provides examples.

4. Embeddings

An embedding is a numerical vector representing the meaning or characteristics of text. Similar meanings tend to have nearby vectors under an appropriate similarity measure. Embeddings enable semantic search beyond exact keyword matching.

5. Vector Databases and FAISS

A vector database stores embeddings and supports similarity search. FAISS is a library for efficient similarity search over dense vectors. A typical process embeds documents, indexes the vectors, and retrieves the nearest vectors for a query.

6. RAG Pipeline

Retrieval-Augmented Generation combines retrieval with text generation. Documents are loaded, split into chunks, embedded, and stored. A user question is embedded; relevant chunks are retrieved; the chunks and question are placed into a prompt; the LLM generates an answer using the supplied context.

7. Grounding and Hallucination

Grounding means connecting an answer to trusted evidence or retrieved context. Hallucination is an unsupported or incorrect generated statement. RAG can reduce hallucinations but cannot guarantee correctness. The system should say when information is missing and display source metadata.

8. Chunking and Retrieval Tuning

Chunk size affects context quality and retrieval precision. Chunk overlap preserves information across boundaries. The number of retrieved chunks is often called k. Reranking can reorder retrieved results using a second relevance model.

9. RAG Evaluation

Evaluate retrieval and generation separately. Retrieval questions include whether relevant passages were found. Generation questions include correctness, relevance, completeness, and groundedness. Use a fixed test set and manually inspect failures.

10. Secure AI Applications

Keep API keys in environment variables, validate uploaded files, limit file sizes, avoid exposing secrets in prompts or logs, handle API failures, and treat retrieved documents as untrusted input. Document the system's limitations.